

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME:

Cloud Computing and
Big Data Analytics

COURSE CODE:

CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I Arjun (192365017) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Arjun Reddy

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1) stand alone computing refers to computers that operate independently using local hardware, software and storage. as the technology evolved, computing shifted to network systems, client-server architecture, distributed computing, virtualization and finally cloud computing.

Key Technology Transitions:

1) stand alone computing: applications and data stored on a single computer.

2) client-server computing: clients access servers from centralized servers.

3) Distributed computing: multiple computers share processing.

4) virtualization: multiple virtual machines run on a single physical server.

5) cloud computing: computing resources are delivered over the internet on demand.

Benefits of cloud computing:

→ scalability.

→ cost efficient.

→ high availability.

→ Remote accessibility.

- 2) A hypervisor is a software that creates and manages virtual machines (VMs). It allows multiple operating systems to run on single physical computers by sharing hardware resources such as CPU, memory and storage.

Type 1 Hypervisors	Type 2 Hypervisors
→ Runs directly on hardware → High performance → more secure → used in data centers - ex: VMware, ESXi	→ Runs on top of host. → slightly slower. → less secure → used for personal computers. - ex: Oracle.

- 3) An API (Application Programming Interface) is set of rules that allows different software applications to communicate.

Role of API in cloud computing:

- connect different cloud services.
- Enable application integration.
- automate cloud operations.
- allows secure data exchange.
- support third-party app development.

Interoperability:
- APIs enable different cloud platforms and applications to exchange data and work together regardless of programming languages or operating system.
ex: mobile app uses an API to store files in google drive.

4) Resource pooling means cloud providers combine computing resources into a shared pool that serves multiple users.

Benefits:

- Efficient utilization of resources.
- Lower infrastructure cost
- Better performance.
- Resources are allocated dynamically based on demand.
- Support multiple users simultaneously without affecting service quality.

ex: many companies use the same AWS data center while each company's data remains isolated and secure.

n) Scalability elasticity is ability of cloud systems to automatically increase & decrease computing resources based on workload demand.

How it supports dynamic workloads:

- adds servers during high traffic

- Remove unused resources during low demand.

- prevents application downtime.

- Improves performance.

- Reduces operational costs by paying only for resources used.

ex: During an online shopping at festival time, cloud servers are automatically scale up to handle millions of users and scale down after the sale end.